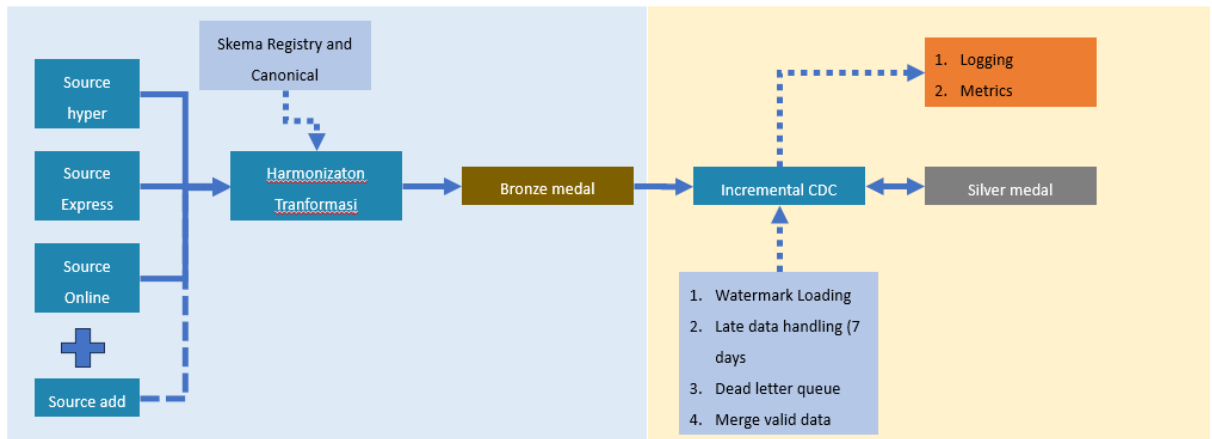


Technical Design Mega Mart

A. Architecture Decisions

1. Pipeline:



2. Fitur:

- Skema harmonization with Skema registry and canonical.
- Watermark loading for merger data validation.
- Dead Letter Queue for invalid data validation.
- Late 7 days data handling for update and new data.

3. Tech Stack used:

- Storage : S3 or Minio using Delta Lake for ACID transactions and CDC Merger operation
- Programing : Python
- Library : [delta-spark==3.2, ipykernel>=7.3.0 , jupyter>=1.1.1 (untuk simulasi), pandas>=3.0.3, pyspark==3.5.1]
- Orchestra : Apache Airflow
- Infrastructure : Docker, Docker Compose, Kubernetes

B. Step Add New Source

- Analyze source schema and business keys.
- Register source metadata in Schema Registry.
- Define source to canonical mapping.
- Configure CDC column and watermark strategy.
- Execute schema harmonization.
- Validate data quality rules.
- Deploy and monitor pipeline execution.

C. Troubleshooting Guide

Issue	Action
Schema Mismatch	Review mapping config from schema registry
Missing Records	Verify Watermark value
Duplicate Records	Check business key and merge condition
Pipeline failure	Check Log in airflow or logging monitor
Invalid Records	Inspect DLQ Records
Slow performance	Review Partitioning and size file

D. Monitoring Approach

1. Records rejected to DLQ.
2. Pipeline duration using airflow.
3. Watermark Refresh.
4. Data volume trend monitoring

“A scalable, metadata driven data platform designed to support multi source integration, incremental CDC processing, and operational monitoring.”